

Quantum Mechanics II

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Introduction

Welcome to Quantum Mechanics II. This course covers important applications of the formalism and foundation you have built in Quantum Mechanics I and explores more advanced topics in quantum theory. We will cover approximation methods, scattering theory, identical particles, and an introduction to relativistic quantum mechanics. For most of our time, we will be concerned with solving the Schrödinger equation with approximate methods. One of the main goals of this course is to equip you with the tools of **perturbation theory** (in a non-relativistic setting). We will have a first look at **relativistic systems** with an approach that is simpler but complementary to what you would see in a quantum field theory course.



Reading: Please note that while I will be providing some typed notes and exercises every now and then, this will be the exception rather than the rule. Instead, expect your main reference to be the textbook for this course (Sakurai and Napolitano). Any assigned reading announcements will be made on ICON.

I will also assign readings from other textbooks that you can find online or in the library to complement Sakurai and Napolitano. One excellent complement is *Principles of Quantum Mechanics* by R. Shankar. Shankar's book is particularly good for those that feel less comfortable with the material from Quantum Mechanics I. If you feel utterly confused about quantum mechanics, I strongly recommend to take a few evenings or a weekend to read all first six chapters of Shankar's book (although some confusion about measurements and their interpretations may never fully go away).

If you feel bored, I recommend Weinberg's *Lectures on Quantum Mechanics* and Feynman's *Lectures on Physics*, Volume III. Later on in the course, you can also check out the Atomic, Molecular, and Optical Physics textbooks by Hertel and Schulz. These books contain many interesting applications of quantum mechanics in the AMO world. You can also read original references by Dirac, Heisenberg, Schrödinger, and John Bell. Please do not hesitate to ask me for more suggestions.

For a full reading list and suggested textbooks, please see the **Reading List on ICON**. I will update the reading list as we progress through the course as necessary.



Exercises: There will be two types of exercises: homework assignments and in-class exercises. Homework assignments are graded and count towards 20% of your final grade. In-class exercises are not graded and are meant to help you understand what we are covering in class and initiate discussions. You are encouraged to work together on both types of exercises. **The more you understand in-class exercises and homeworks, the more prepared you will be for the exams.**



Calendar: Below you will find some important dates and a tentative calendar of topics we will cover throughout the semester. This is also available in the course **syllabus** (available on ICON).

Important Dates

- 1st Midterm: Friday, March 6th, 2026
 - 2nd Midterm: Friday, April 24th, 2026
 - Final exam: Monday, May 11th, 2026
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Tentative Topics Calendar

- **Module 1: Review and WKB**
 - *Topics:* Solutions to Schrödinger's equation, Dirac notation and wavefunctions, time-evolution, and the WKB approximation.
 - *Dates:* Weeks 1–2 (Jan 19–30)
 - *Note:* MLK day (No class Jan 19)
- **Module 2: Perturbation Theory**
 - *Topics:* Time-independent and time-dependent perturbation theory. Examples and degenerate cases. Sudden and adiabatic limits of time-dependent problems. Dyson series and applications.
 - *Dates:* Weeks 3–6 (Feb 2–27)
- **Module 3: Scattering Theory**
 - *Topics:* Rates and cross sections, the Born approximation, partial waves, and resonances.
 - *Dates:* Weeks 7–10 (Mar 2–27)
 - *Note:* Spring Break during Week 9 (Mar 16–20)
- **Midterm 1:** Friday, March 6, 2026
- **Module 4: Identical Particles**
 - *Topics:* Fermions and bosons, multiparticle systems, and a first look at second quantization.
 - *Dates:* Weeks 11–12 (Mar 30–Apr 10)
- **Module 5: Relativistic Quantum Mechanics**
 - *Topics:* Relativistic systems and the Klein-Gordon and Dirac equations. Spinors, Dirac algebra, and particle-antiparticle interpretation.
 - *Dates:* Weeks 13–15 (Apr 13–May 1)
- **Midterm 2:** Friday, April 24, 2026
- **Overflow & Review**
 - *Dates:* Week 16 (May 4–8)
- **Final Exam:** Monday, May 11, 2026

Review Resources

In the first two weeks of class we will review some important concepts from Quantum Mechanics I that will be useful throughout the course.

You should feel comfortable with **Dirac notation** and **time-evolution** of quantum states. We will also briefly review the **WKB approximation**, mostly because this is a useful example of an approximation method that is widely used in quantum mechanics, but that is very different from the kinds of approximations that we will study later in the course (perturbation theory).



- **Dirac notation and wavefunctions:**

- Sakurai and Napolitano, Sections 1.2, 1.6, and 1.7. These must become second nature to you.
- Additional reading: Shankar, Chapter 4 and 6 (you can read these entirely).

- **Time-evolution:**

- Sakurai and Napolitano, Sections 2.1 and 2.4.
- Schrödinger and Heisenberg pictures: Sakurai and Napolitano, Section 2.2.
- See also the propagator approach in Sections 2.6 and the density matrix approach in Section 3.4.
- Additional reading: Shankar, Section 4.3.

- **Standard Problems:**

- Sakurai and Napolitano, parts of Section 2.5 but mostly the elementary problems in this chapter (free particle case, finite and infinite potential wells, and the harmonic oscillator).
- Additional reading: Shankar, Chapter 5. If you understand “Section 5.4 - A Problem in Scattering” you are in good shape (we will do this in class and later revisit it with more machinery).